

21 DAYS POWER BI DASHBOARD CHALLENGE

TASK 10 - DOCUMENTATION & SUBMISSION

1. About the Dataset

Dataset Name: Sales / Order Details Dataset

Order ID	Order Date	CustomerName	State	City
B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad
B-25602	01-04-2018	Pearl	Maharashtra	Pune
B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal
B-25604	03-04-2018	Divsha	Rajasthan	Jaipur
B-25605	05-04-2018	Kasheen	West Bengal	Kolkata
B-25606	06-04-2018	Hazel	Karnataka	Bangalore
B-25607	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir
B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai
B-25609	09-04-2018	Jitesh	Uttar Pradesh	Lucknow
B-25610	09-04-2018	Yogesh	Bihar	Patna
B-25611	11-04-2018	Anita	Kerala	Thiruvananthapuram
B-25612	12-04-2018	Shrichand	Punjab	Chandigarh
B-25613	12-04-2018	Mukesh	Haryana	Chandigarh
B-25614	13-04-2018	Vandana	Himachal Pradesh	Simla
B-25615	15-04-2018	Bhavna	Sikkim	Gangtok
B-25616	15-04-2018	Kanak	Goa	Goa
B-25617	17-04-2018	Sagar	Nagaland	Kohima
B-25618	18-04-2018	Manju	Andhra Pradesh	Hyderabad

Dataset Description

The dataset contains transactional sales data with details about customer orders, product categories, revenue, quantity sold, and profit. It is designed to analyze sales performance, profitability, and trends over time.

Key Tables Used

- **Order Details** (Fact Table)
- **Sales Target** (Target / Reference Table)
- **Date Table** (Created using Order Date hierarchy)

Key Columns

- Order ID
- Order Date
- Category
- Sub-Category
- State
- Amount (Sales Value)
- Quantity
- Profit

2. Data Cleaning Steps

Data cleaning was performed using **Power Query Editor** in Power BI.

Steps Followed

1. Removed duplicate records from Order ID column
2. Checked and corrected data types:

Amount → Decimal Number

Profit → Decimal Number

Quantity → Whole Number

Order Date → Date

3. Removed null and blank values from key columns
4. Standardized Category and Sub-Category names
5. Ensured consistent formatting for State names
6. Verified negative profit values to reflect loss-making orders
7. Created calculated columns where required for analysis

After cleaning, the dataset was loaded into Power BI for modeling and visualization.

The screenshot displays the Power Query Editor interface. The main area shows a table with 16 rows of order data. The columns are: Order ID, Order Date, CustomerName, State, City, and Location. The 'Location' column contains the state and city names. The 'Applied Steps' pane on the right shows the sequence of transformations applied to the data, including 'Source', 'Use First Row as Headers', 'Change Type', 'Removed Duplicates', 'Capitalized Each Word', 'Added Custom', and 'Capitalized Each Word1'.

Order ID	Order Date	CustomerName	State	City	Location
1	01-04-2018	Bharat	Gujarat	Ahmedabad	Ahmedabad, Gujarat
2	01-04-2018	Pearl	Maharashtra	Pune	Pune, Maharashtra
3	03-04-2018	Jahan	Madhya Pradesh	Bhopal	Bhopal, Madhya Pradesh
4	03-04-2018	Divsha	Rajasthan	Jaipur	Jaipur, Rajasthan
5	05-04-2018	Kasheen	West Bengal	Kolkata	Kolkata, West Bengal
6	06-04-2018	Hazel	Karnataka	Bangalore	Bangalore, Karnataka
7	06-04-2018	Sonakshi	Jammu And Kashmir	Kashmir	Kashmir, Jammu And Kashmir
8	08-04-2018	Aarushi	Tamil Nadu	Chennai	Chennai, Tamil Nadu
9	09-04-2018	Jitesh	Uttar Pradesh	Lucknow	Lucknow, Uttar Pradesh
10	09-04-2018	Yogesh	Bihar	Patna	Patna, Bihar
11	11-04-2018	Anita	Kerala	Thiruvananthapuram	Thiruvananthapuram, Kerala
12	12-04-2018	Shrichand	Punjab	Chandigarh	Chandigarh, Punjab
13	12-04-2018	Mukesh	Haryana	Chandigarh	Chandigarh, Haryana
14	13-04-2018	Vandana	Himachal Pradesh	Simla	Simla, Himachal Pradesh
15	15-04-2018	Bhavna	Sikkim	Gangtok	Gangtok, Sikkim
16	15-04-2018	Kanak	Goa	Goa	Goa, Goa

3. Data Model

Data Model Structure

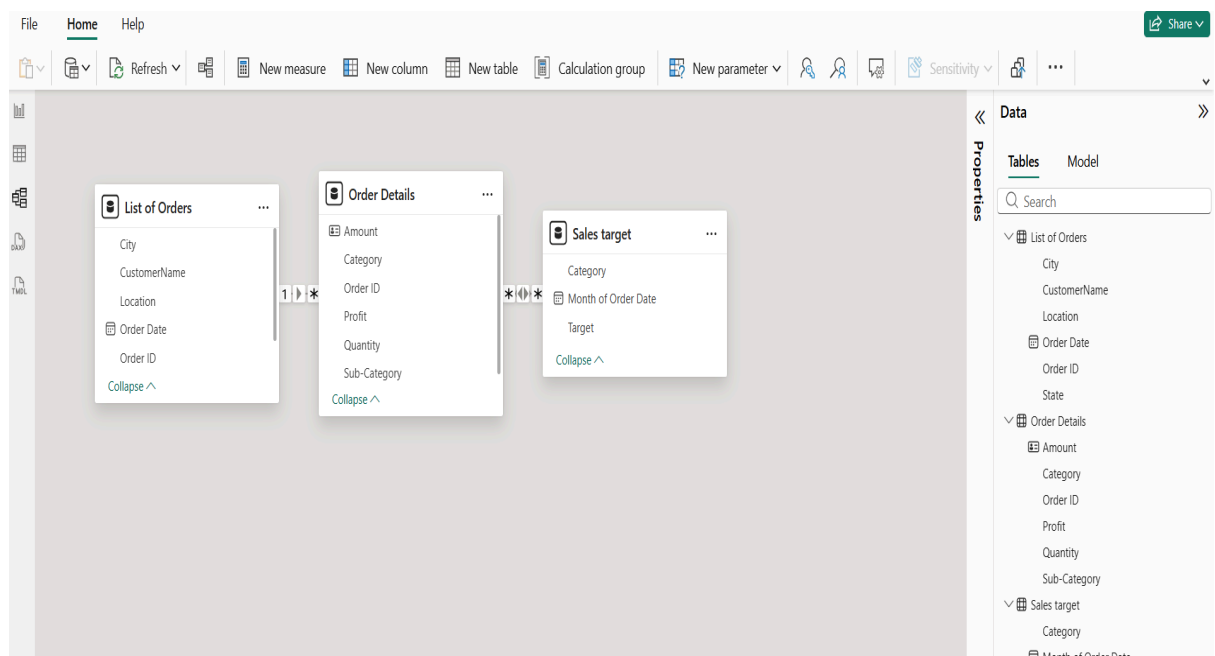
- **Order Details** acts as the **Fact Table**
- **Date Table** acts as the **Dimension Table**
- **Sales Target** linked for performance comparison

Relationships

- Order Details[Order Date] → Date Table[Date] (Many-to-One)
- Category/Sub-Category used for slicing and filtering

Model Highlights

- Proper cardinality maintained (One-to-Many)
- Cross filter direction set to **Single** for better performance
- Star schema followed for optimized reporting



4. DAX Measures Created

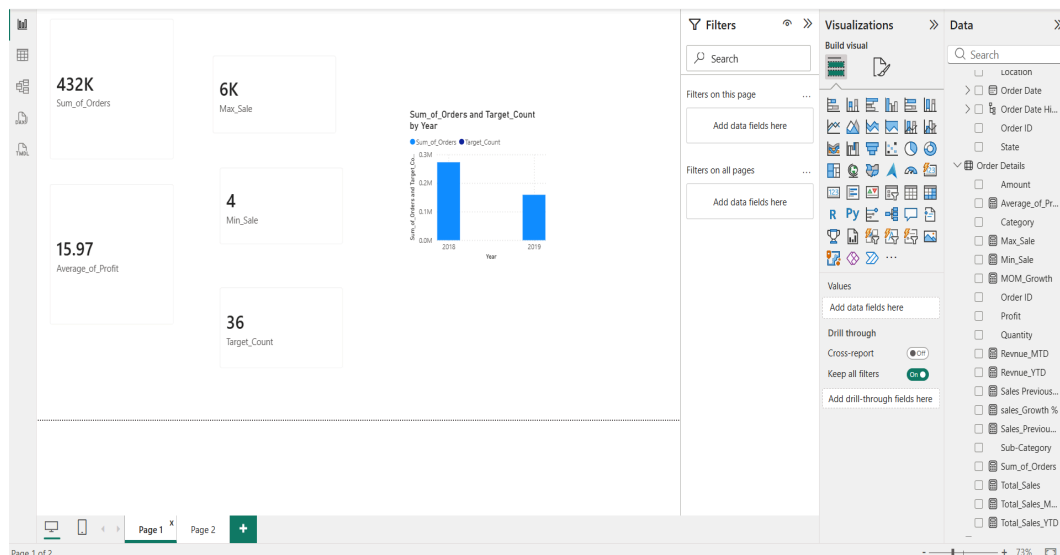
Basic Measures

Total Sales = SUM('Order Details'[Amount])

Total Profit = SUM('Order Details'[Profit])

Total Quantity = SUM('Order Details'[Quantity])

Total Orders = DISTINCTCOUNT('Order Details'[Order ID])



Time Intelligence Measures

Sales Previous Month =

```
CALCULATE(  
    [Total Sales],  
    PREVIOUSMONTH('Date Table'[Date])  
)
```

Sales Previous Year =

```
CALCULATE(  
    [Total Sales],  
    SAMEPERIODLASTYEAR('Date Table'[Date])  
)
```

Revenue MTD = TOTALMTD([Total Sales], 'Date Table'[Date])

Revenue YTD = TOTALYTD([Total Sales], 'Date Table'[Date])

Advanced Measures

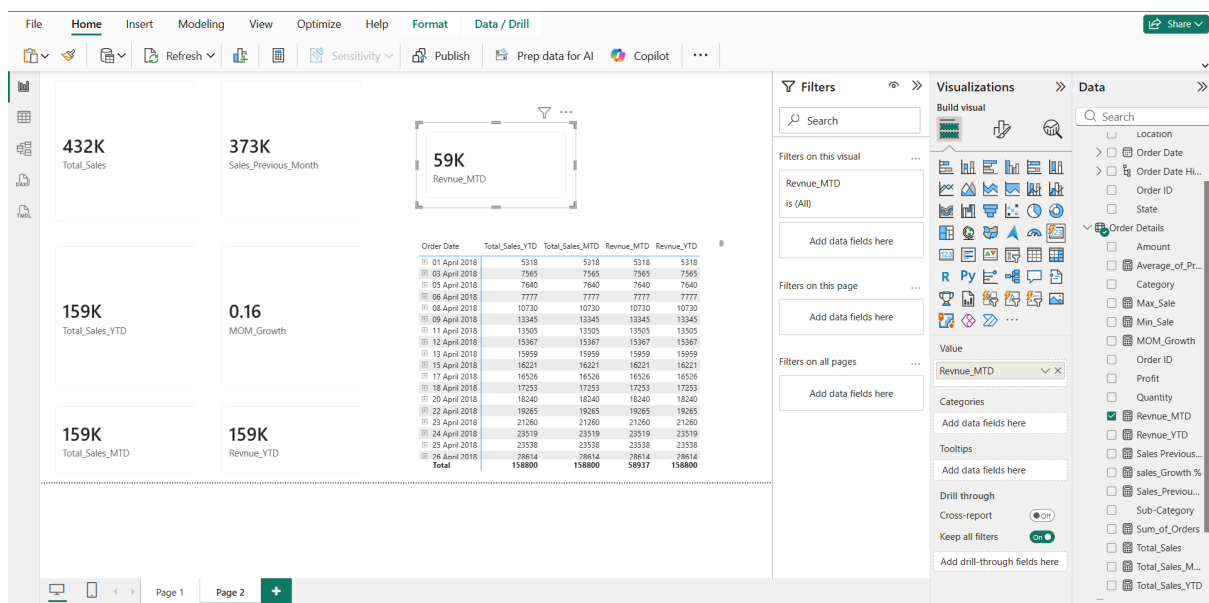
MoM Growth % =

DIVIDE([Total Sales] - [Sales Previous Month], [Sales Previous Month])

Average Profit = AVERAGE('Order Details'[Profit])

Max Sale = MAX('Order Details'[Amount])

Min Sale = MIN('Order Details'[Amount])



5. Insights Discovered

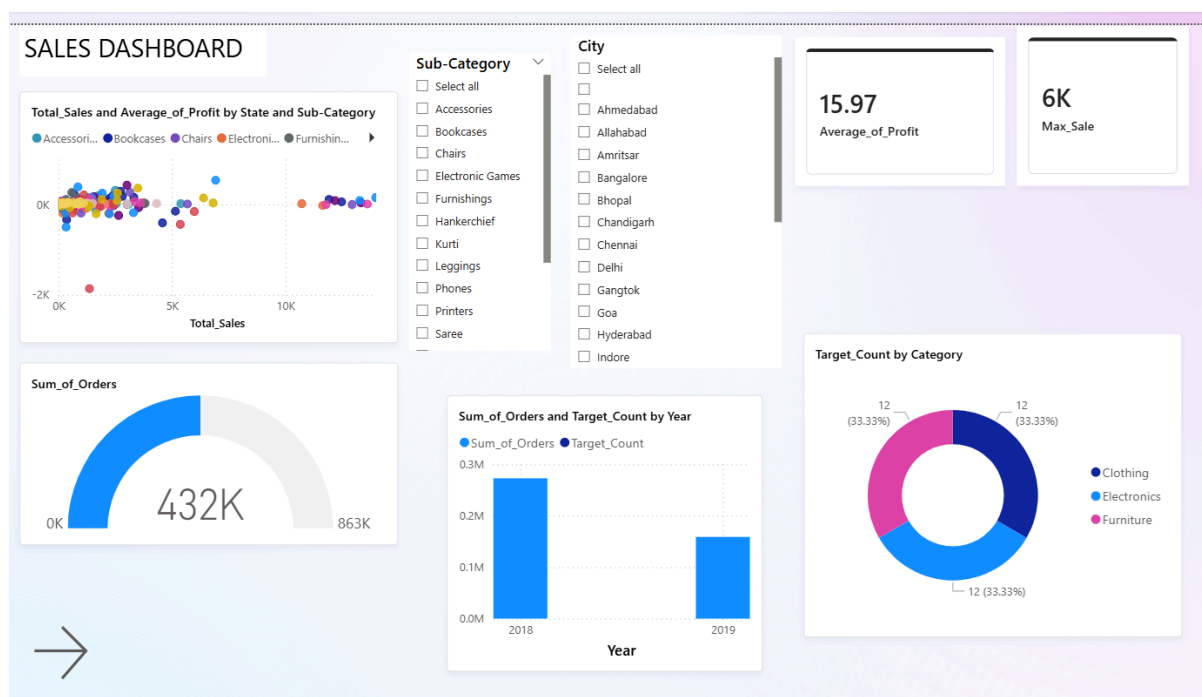
Key Business Insights

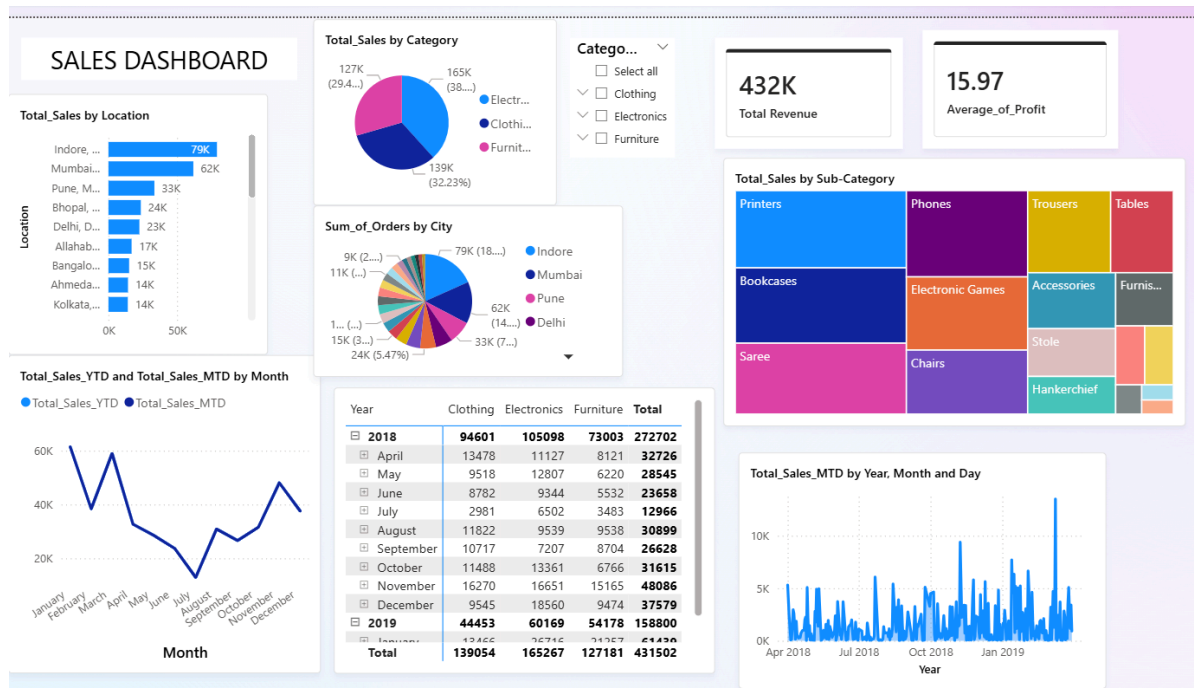
- Clothing category contributes the highest number of orders
- Saree sub-category shows high sales volume but fluctuating profit
- Several orders generate negative profit, indicating discount or cost issues
- Month-on-Month growth highlights seasonal demand patterns
- Certain states outperform others consistently in total revenue
- Sales targets are met in peak months but missed during off-season periods

Dashboard Highlights

- KPI Cards for Total Sales, Profit, Quantity, Orders
- Bar charts for Category and Sub-Category performance
- Line chart for monthly sales trend
- Table visual for detailed order-level analysis
- Slicers for Category, Sub- Category, City

6. Final Dashboard





7. Conclusion

This project demonstrates the complete Power BI workflow including data cleaning, modeling, DAX calculations, and dashboard creation. The analysis provides actionable insights into sales performance, profitability, and growth trends, making it useful for business decision-making.